

Puneeth Naidu

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Summary

Aspiring Machine Learning Engineer with a Master's degree in Computer Science and strong knowledge in Artificial Intelligence, Machine Learning, Deep Learning, and Data Analytics. Experienced in developing AI/ML academic and personal projects involving recommendation systems, Generative AI, Retrieval-Augmented Generation (RAG), NLP, and predictive analytics. Skilled in Python, SQL, FastAPI, TensorFlow, Scikit-learn, and data processing technologies. Passionate about building intelligent systems, solving real-world problems, and continuously learning emerging AI technologies.

Technical Skills

- **Programming Languages:** Python, SQL
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost
- **Generative AI & LLMs:** LangChain, RAG, Prompt Engineering, FAISS
- **Backend Development:** FastAPI, Flask, REST APIs
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Data Processing:** Pandas, NumPy, PySpark
- **Tools & Platforms:** Git, Docker, Linux, Jupyter Notebook
- **Visualization:** Power BI, Tableau, Matplotlib

Academic Projects

AI Resume Screening System

- Developed an AI-powered resume screening application using Python, FastAPI, and NLP techniques.
- Implemented ATS-style resume matching, skill extraction, candidate ranking, and interview question generation.
- Integrated Generative AI models for resume analysis and hiring recommendations.

Local RAG Chatbot

- Built a Retrieval-Augmented Generation chatbot using LangChain, FAISS, and Ollama.
- Implemented document ingestion, vector embeddings, semantic search, and contextual response generation.
- Enabled querying of PDF documents using local Large Language Models.

Movie Recommendation System

- Developed a content-based recommendation system using TF-IDF and cosine similarity.
- Implemented movie retrieval and recommendation workflows using Python and Scikit-learn.

Face Mask Detection System

- Built a deep learning-based face mask detection system using TensorFlow, Keras, and OpenCV.
- Implemented real-time image classification using Convolutional Neural Networks.

Education

Master of Science in Computer Science

DePaul University, Chicago, Illinois

Graduated: June 2025

Experience

Client: AIM BIG SOLUTIONS ,florida

July 2025 – Present

Role: AI/ML Engineer

- Developed patient risk prediction and hospital readmission models using clinical and claims datasets to support proactive healthcare decision-making and improve care management workflows.
- Utilized Python and PySpark to process, transform, and analyze large-scale structured and unstructured healthcare datasets from multiple enterprise data sources.
- Engineered temporal and longitudinal healthcare features including visit frequency, chronic condition indicators, and rolling aggregations to improve predictive model accuracy and reliability.
- Trained and optimized XGBoost, Logistic Regression, and Neural Network models for healthcare risk scoring and predictive analytics use cases.
- Deployed ML models on AWS SageMaker and integrated FastAPI-based REST APIs to support scalable real-time inference workflows in production environments.
- Implemented MLflow-based model monitoring, logging, and data drift detection workflows to improve model reliability and operational stability.
- Built semantic search and healthcare document retrieval pipelines using vector embeddings, FAISS, and LangChain for knowledge discovery applications.

Environment: Python, SQL, PySpark, TensorFlow, PyTorch, AWS (SageMaker, S3, EMR, Bedrock), LangChain, FAISS, Kafka, MLflow, Docker, FastAPI, GitHub Actions, Git, Linux

Client: LTTS, India

Feb 2021 - Aug 2022

Role: software Engineer

- **Built scalable PySpark pipelines to process large-scale IoT and industrial sensor data from connected devices and enterprise monitoring systems.**
- **Processed high-volume time-series data using Apache Spark to support real-time equipment monitoring, anomaly detection, and operational analytics workflows.**
- **Developed ETL workflows to ingest, transform, and integrate data from multiple structured and streaming data sources for downstream analytics and reporting.**
- **Created feature engineering pipelines for predictive maintenance and equipment failure analysis using historical sensor and operational datasets.**
- **Optimized SQL queries, Spark jobs, and data transformation workflows to improve processing efficiency and reduce execution time across distributed systems.**
- **Worked with large-scale datasets using Hive and Spark to support distributed data processing and analytics applications in production environments.**
- **Collaborated with data scientists and analytics teams to deliver curated datasets and engineered features for machine learning model development.**
- **Automated batch data processing and validation workflows to improve data quality, pipeline reliability, and operational efficiency.**
- **Participated in Agile development activities including sprint planning, code reviews, deployment support, and cross-functional collaboration.**

Environment: Python, SQL, PySpark, Pandas, NumPy, Apache Spark, Hive, Kafka, ETL Pipelines, Data Warehousing, Docker, Git, Linux

Strengths

- Strong analytical and problem-solving skills
- Passionate about Artificial Intelligence and Machine Learning
- Quick learner with adaptability to new technologies
- Good communication and teamwork abilities
- Strong foundation in programming and software development